F78PA Probability and Statistics A _Final Exam Semester 1 2018/19

Answers

1. (a) See lecture notes

   (b) $0 \leq c \leq \frac{2}{3}$, this will give a function which is non-negative and right continuous with left limits.

   (c) F_X is a continuous function and differentiable almost everywhere, so X has a density function.

   (e) $\frac{8}{9}$

   (f) (ii) $f_Y(y) = \begin{cases} \frac{2}{3}\left(\frac{1}{y^2} + \frac{1}{y^3}\right), & y \geq 1 \\ 0, & \text{otherwise} \end{cases}$

2. (b)

$$f_Y(y) = \begin{cases} \frac{3}{8}(2-x)^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

   (i) $\frac{4}{9}$ (ii) 1

(c) ImZ = [0, 2] , $F_Z(z) = \begin{cases} 0 & \text{if } z < 0 \\ \frac{z^3}{8} & \text{if } 0 \leq z \leq 2 \\ 1 & z > 2 \end{cases}$, $f_Z(z) = \begin{cases} \frac{3}{8}z^2, & 0 \leq z \leq 2 \\ 0, & \text{otherwise} \end{cases}$

3. (a) (ii) $F_T(t) = \begin{cases} 0 & \text{if } t < 0 \\ \left(\frac{t}{15}\right)\left(1 - e^{-\frac{t}{10}}\right) & \text{if } 0 \leq t \leq 15 \\ 1 - e^{-\frac{t}{10}} & \text{if } t > 15 \end{cases}$, 0.818

   (b) Bill needs to leave at 7.58am.

4. (b) $Y \sim \Gamma(\alpha, \frac{\lambda}{c})$

6. (b) 1.38 (d) 0.3803